which is exactly the classical relation between coordinates and their conjugate momenta (see *Volume I*, Lecture 10, Eq. 8). Ultimately, it is this connection that explains why the quantum concept of momentum is connected to the classical concept.

Using the general uncertainty principle from Lecture 5, we can now specialize to the case

$$[\mathbf{X}, \mathbf{P}] = i\hbar.$$

and

$$\Delta\mathbf{X}\Delta\mathbf{P} \geq \frac{\hbar}{2}.$$

We'll do that in the next section.

Now let's recall the second principle involving commutators. In Lecture 4, we found that two observables **L** and **M** cannot be determined simultaneously unless they commute. If they don't commute, you cannot measure **L** without interfering with a measurement of **M**. It is not possible to find simultaneous eigenvectors of two noncommuting observables. This led to the general uncertainty principle.

8.5 THE HEISENBERG UNCERTAINTY PRINCIPLE

And now, ladies and gentlemen, here's what you've all been waiting for. At long last: *the Heisenberg Uncertainty Principle*.

The Heisenberg Uncertainty Principle is one of the most famous results of quantum mechanics: it not only asserts that the position and momentum of a particle cannot be simultaneously known, but it also provides an exact quantitative limit for their mutual uncertainties. At this point, I suggest that you revisit Lecture 5, where I explained the general uncertainty principle. We did all the work there, and now we get to reap the benefits.